

Academic Excellence Through Quality Education
KATHMANDU MODEL SECONDARY SCHOOL

FIRST TERM EXAMINATION 2082

Set: XI 'Science'

SET 'A'

Set: 2 hrs.

Subject: Computer Science (4271)

F.M.: 50

P.M.: 20

(Candidates are required to give their answer in their own words as far as practicable. To figure in the margin indicated full marks.)

Attempt all the questions.

GROUP 'A' [9×1=9]

MULTIPLE CHOICE QUESTIONS.

1. _____ are numbers and encoded characters generally used as operands.
 a. Input b. Data c. Information d. Stored values
2. Analog Computer works on the supply of:
 a. Physical strength b. Discrete signals
 c. Magnetic strength d. Continuous electric pulse
3. Which is the major component of first generation computers?
 a. Transistor b. Microprocessor
 c. Vacuum Tube d. IC
4. A bar code reader is an example of _____ device.
 a. processing b. storage c. Input d. output
5. What is the binary equivalent of a decimal number 21?
 a. 10101 b. 10100 c. 11010 d. 101101
6. Which of the following isn't positional number system?
 a. Binary b. Octal c. Decimal d. Roman
7. A light sensitive device that converts drawings, printed text or other images into digital form is
 a. Keyboard b. Scanner c. Plotter d. OMR
8. A magnetic tape is a type of _____ access storage device.
 a. Sequential b. Direct c. Step d. Indirect
9. What is the maximum Zoom size in MS word?
 a. 150 b. 200 c. 500 d. 350

GROUP 'B' [5×5=25]

SHORT ANSWER TYPE QUESTIONS.

10. Why we use multimedia? Define linear and non-linear multimedia with examples. Explain the different components of multimedia. [1+1+3]
11. Explain the memory hierarchy in a computer system with a neat and labeled diagram. Describe the different levels of memory and discuss their characteristics in terms of speed, cost, and capacity. [2.5+2.5]
12. Describe any three types of positional number system. Convert the following as indicated. [3+1+1]

a. $(93.6)_{10} = (?)_2$

b. $(BAD.DAD)_{16} = (?)_{10}$

'OR'

Subtract 1011_2 from 10101_2 using 1's and 2's complement methods.

[2.5+2.5]

13. What is mobile computing? Suppose the internet and mobile networks suddenly stopped working for one full week. Describe at least two problems people would face in education, business, and daily life. Suggest at least two alternative solutions people could use during that time. [1+1+1+1+1]

'OR'

Explain in detail the concept of mail merge in Microsoft Word. What are its primary benefits and disadvantages? [5]

14. Define output devices with examples. Write their functions. Write any four differences between CRT monitor and LCD monitor. [1+2+2]

GROUP 'C' [2×8=16]

LONG ANSWER TYPE QUESTIONS.

15. What do you mean by computer architecture? Describe each functional components of computer architecture with typical diagram. [1+7]
16. You are given to prepare a project report in MS Word about "Generations of Computers."
 a. Explain how you would use any four Word features (like Table of Contents, Page Numbering, WordArt, Styles, SmartArt, etc.) to organize the report neatly. [4×1.5=6]
 b. Write why these features are useful for readers. [2]

"Best of Luck"

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GROUP 'A' [9×1=9]

MULTIPLE CHOICE QUESTIONS.

1. The three main components of computer are
 a) CPU, Memory, Tape b) I/O, Printer, Mouse
 c) Tape, I/O, Floppy disk d) CPU, I/O, Memory
2. Which type of computer is smaller than a mainframe but larger than a microcomputer, and used by medium-sized organizations?
 a) Hybrid Computers b) Super Computers
 c) Mini Computers d) Analog Computers
3. Which is the major component of fourth generation computers?
 a) Transistor b) Microprocessor c) Vacuum Tube d) IC
4. A speaker is an example of device.
 a) processing b) storage c) input d) output
5. What is the binary equivalent of a decimal number 31?
 a) 10101 b) 11111 c) 11010 d) 101101
6. How many bits are there in 2.5 kilobytes?
 a) 20,480 bits b) 25,600 bits c) 40,960 bits d) 20,480 bytes
7. The maximum number of bits used to represent hexadecimal number in binary is
 a) 3 b) 4 c) 8 d) 16
8. The _____ is a small input device used to select and display objects on a screen.
 a) Ink b) Magnet c) Light pen d) None of these
9. Which of the following is not the font style?
 a) Bold b) Italics c) Regular d) Superscript

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12. Describe any three types of positional number system. Convert the following as indicated. [3+1+1]

a) $(93.6)_{10} = (?)_2$

b) $(BAD.DAD)_{16} = (?)_8$

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Subtract 1011_2 from 10101_2 using 1's and 2's complement methods. [2.5+2.5]

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GROUP 'A' [9×1=9]

Multiple Choice Questions.

1. The term GIGO is related to which characteristics of computers?
 - a) Speed
 - b) Automatic
 - c) Accuracy
 - d) Reliability
2. **NAND gate means**
 - a) Inversion followed AND gate
 - b) AND gate followed by an inverter
 - c) AND gate followed by an OR gate
 - d) None of above
3. Which software is used to help analyze, configure, optimize and maintain the computer?
 - a) Device driver
 - b) Utility software
 - c) Application software
 - d) All
4. Which symbol must all formula begin within MS-Excel?
 - a) =
 - b) +
 - c) %
 - d) @
5. The numeric data entered in the worksheet is _____ aligned by default.
 - a) Left
 - b) Right
 - c) Center
 - d) Top
6. Which HTML tag is used to create smallest heading in a webpage?
 - a) <h0>
 - b) <h3>
 - c) <h6>
 - d) <h7>
7. **JPEG stands for _____.**
 - a) Joint photographic Exports Group
 - b) Joint Physical Exports Group
 - c) Joint Physical Exports Group
 - d) Joint Photographic Exports Group
8. **Microsoft Windows is an example of?**
 - a) Open source OS
 - b) Proprietary OS
 - c) Both
 - d) None
9. **Programs stored in the ROM are called as:**
 - a) Firmware
 - b) Formware
 - c) Farmware
 - d) Firmware

GROUP 'B' (5×5=25)

Short Answer Questions.

10. Justify the statement: "The mechanical nature of forming characters means that **impact printers** generate much higher noise levels during operation, which makes them unsuitable for quiet environments where **non-impact printers** are the preferred choice."
11. Explain the primary memory with its type. [5]
12. Differentiate between **System Software** and **Application**.

'OR'

Differentiate between Linear Multimedia and Non-Linear Multimedia.

13. Write HTML code to print following:
- a. First Generation
 - i. Vacuum Tube
 - ii. Machine Language
 - iii. ENIAC
 - b. Second Generation
 - i. Transistors
 - ii. Assembly Language
 - iii. IBM 1401
 - c. Third Generation
 - i. Integrated Circuit
 - ii. High Level Language
 - iii. PDP-8
14. In Excel, why we use Cell referencing? Explain Absolute Cell Referencing and Relative Cell Referencing.

GROUP 'C' [2×8=16]

Long Answer Questions.

15. What do you mean by Operating system? Explain any 6 types of operating system on the basis of **Processing Mode**. [2+6]
16. Explain all Logic Gates with **statements, Boolean expression, logical symbol, truth table and Venn-diagram**.

Best of Luck



KATHMANDU MODEL SECONDARY SCHOOL

SECOND TERM EXAMINATION 2082

Class: XI 'Science'

SET 'A'

F.M.: 50

Time: 2 hrs.

Subject: Computer Science (4271)

P.M.: 20

Candidates are required to give their answer in their own words as far as practicable.

The figure in the margin indicated full marks:

Attempt all the questions.

GROUP 'A' [9×1=9]

Multiple Choice Questions.

1. Storage which stores or retains data after power off is called _____.
a) Volatile storage b) Non-volatile storage
c) Cache memory d) None of above
2. Which logic gate is represented by the Boolean function $F = A \cdot B + A' \cdot B$?
a) XOR b) XNOR c) AND d) OR
3. Which software is used to perform specific task?
a) System software b) Application software
c) Utility software d) None
4. Which one of the following font-size can not be selected from tool bar?
a) 11 b) 12 c) 13 d) 14
5. What is the keyboard shortcut to start a PowerPoint presentation from the beginning?
a) F5 b) Ctrl + F5 c) Alt + F5 d) Shift + F5
6. Which of the following tags do you use to create an ordered list?
a) `<a href="">` b) `<UL>` c) `<OL>` d) `<LIST>`
7. A video consists of a sequence of _____.
a) Signals b) frames c) packets d) slots
8. Linux, Unix etc. are the examples of?
a) Proprietary OS b) Open source OS
c) Both d) None

9. Which type of memory can speed up computer processing?
a) ROM b) Cache memory
c) Registers d) Both A and B

GROUP 'B' [5×5=25]

Short Answer Questions.

10. Justify the statement: "The mechanical nature of forming characters means that impact printers generate much higher noise levels during operation, which makes them unsuitable for quiet environments where non-impact printers are the preferred choice." [5]
11. Explain the primary memory with its type. [5]
12. Differentiate between System Software and Application Software. [5]

'OR'

Differentiate between Linear Multimedia and Non-Linear Multimedia. [5]

13. Write HTML code to print following:

- a. First Generation
 - i. Vacuum Tube
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- b. Second Generation
 - i. Transistors
 - ii. Assembly Language
 - iii. IBM 1401
- c. Third Generation
 - i. Integrated Circuit
 - ii. High Level Language
 - iii. PDP-8

14. In Excel, why we use Cell referencing? Explain Absolute Cell Referencing and Relative Cell Referencing. [1+4]

GROUP 'C' [2×8=16]

Long Answer Questions.

15. What do you mean by Operating system? Explain any 6 types of operating system on the basis of Processing Mode. [2+6]
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